Homework 8

PHYS 6020 - Fall 2023

Due on Monday, November 3rd, 2023

Review the solved problems at the end of Chapter 5 of Zettili's book

Problem 1

In addition to the basis defined by its eigenstates $|n\rangle$, with $n = 0, 1, 2, \cdots$, the single harmonic oscillator Hamiltonian, $H_{HO} = \hbar\omega(n + 1/2)$, can be written in terms of the basis of coherent states $\{|\alpha\rangle\}$. These are defined as eigenstates of the lowering (annihilation) operator (here α is a complex number):

$$\hat{a}|\alpha\rangle = \alpha|\alpha\rangle$$

(i) Show that

$$|\alpha\rangle = e^{\alpha\hat{a}^\dagger}|0\rangle$$

where $|0\rangle$ is the ground state of the harmonic oscillator. Hint: remember the definition for the exponential of an operator and use the properties of commutators.

(ii) Calculate the probability of getting a value for the energy $E_n = \hbar\omega(n + 1/2)$ in the coherent state $|\alpha\rangle$. The result is a function of α. Sketch this function.

(iii) Calculate the mean value of the number operator $\langle\alpha|\hat{N}|\alpha\rangle$, and the Hamiltonian operator $\langle\alpha|\hat{H}|\alpha\rangle$ and their dispersions $\Delta\hat{N}$ and $\Delta\hat{H}$. What are the ratios $\Delta\hat{N}/\langle\alpha|\hat{N}|\alpha\rangle$, and $\Delta\hat{H}/\langle\alpha|\hat{H}|\alpha\rangle$. What do you conclude?

Problem 2

Solve problem 5.4 at the end of Chapter 5 of Zettili's book.

Problem 3

Solve problem 5.10 at the end of Chapter 5 of Zettili's book. Do only part a).

Problem 4

Solve problem 5.11 at the end of Chapter 5 of Zettili's book.

Problem 5

Solve problem 5.16 at the end of Chapter 5 of Zettili's book.

Problem 6

Solve problem 5.18 at the end of Chapter 5 of Zettili's book.